

Discussion on Testing

17-316/616 Fall 2025

AI Tools for Software Development

<https://ai-developer-tools.github.io>

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Dispatches from ASSETS 2025

- There were several cool papers about using LLMs to support people with disabilities.
 1. Fixing WCAG errors in web pages.
 2. Checking PDF accessibility
 3. Improving CART + ASR accuracy
 4. Bridging communication misunderstandings between autistic and non-autistic people.

P5 Questions

- What questions do you have about P5: Testing?

Reflection prompts this week

- How can you verify that the tests generated by the LLM are sufficient? What do you need in order to feel confident in the test cases (and why)? What do you think of the paper's remarks on this? When do you stop testing?
- Do you believe it is effective to use AI to test code generated by an AI? If hallucinations can happen with the original writing of the code, will that be compounded by the writing of the tests? How involved do you think human software developers need to be and why?
- Imagine that you are the VP of Engineering for a software startup. You need to define the company's policy on testing. Write the policy on how your software developers should approach automated testing, when to test, how much to test, how many tests should be written, how to involve AI, etc. Explain the rationale for your policy to the CEO. Remember that you are responsible for reliably managing your customers' data, so they don't sue your fledgling company out of business.
- What are the problems that AI has in generating effective unit tests? List 3 problems that have occurred in your experience, and list 3 prompts that you used to solve them.
- How do you get the LLM to keep tests up to date as you make changes to the program code? What would you add to your prompt to ensure the code and tests are updated together? Test this prompt and tell us how it went. Make a change to that prompt that results in an improvement and tell us how it went.
- Make the LLM generate code to pass the unit tests from Wednesday's mobbing session involving the Tinder app. Run the code. How close did this get to Tinder? What is missing? Describe what you believe the additional tests you would need to get it to create the rest of Tinder.

Discussion Instructions

- Instructor announces the reflection question.
- If you chose this reflection question to answer for homework, come up to the front. You are the discussant group.
- Class forms into pairs.
- Pairs and discussant group should now reflect on the question by themselves (4 min)
 - Explain what you think, why, and how you've come to your conclusions.
- Each discussant explains their answer to the class (3-4 min)
- Instructor will choose a pair to report their answer to the class (1 min)
- A discussant will respond and connect the answer to their own reflection (1 min)

Reflection Prompt

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Reflection Prompt

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Next Class

- Deploy your Apps on AWS!
- This is hard, even when we know what to do.
- Can LLMs lift the burden?